Abeer Hashmi

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PROFESSIONAL SUMMARY

Engineering Physics student with hands-on experience in mechanical design, SolidWorks CAD, and physical prototyping through personal projects, design teams, and co-op experience.

EDUCATION

McMaster University — Bachelor of Engineering (Co-op), Engineering Physics *Hamilton, ON • 2025 – Present*

TECHNICAL SKILLS

CAD & Design: SolidWorks, mechanical component design, engineering drawings, BOM development
Mechanical Engineering: GD&T, tolerancing, machining decision matrices, material selection
Prototyping & Manufacturing: 3D printing, machining, physical prototyping, mechanical assembly
Documentation: BOM development, technical drawings, SOPs, technical documentation

PROJECTS

3D-Printed Warehouse Loading Gripper *Personal Project*

- 3D printed a robotic gripper to load and handle warehouse products, applying DFMA to reduce part count and simplify assembly
- Used decision matrices to evaluate gripper mechanisms and select materials based on strength-to-weight and cost trade-offs
- Applied GD&T and tolerancing to mating components for reliable fit and consistent grip performance
- Iterated through multiple prototypes to test and verify grip strength and product compatibility

Battery-Operated Handheld Fan *Personal Project*

- 3D printed a battery-powered handheld fan, developing SolidWorks CAD models and engineering drawings
- Applied a decision matrix to select motor, battery, and housing components based on power, weight, and runtime
- Toleranced and assembled components for proper fit between housing, electronics, and blade assembly

ENGINEERING EXPERIENCE

McMaster Pumpkin Chunkin — Junior Power (Design Team) *Hamilton, ON • 2026 – Present*

- Modeled a mounting system, chain tensioner, and sprockets in SolidWorks using GD&T and tolerancing for proper fit
- Produced BOMs and engineering drawings to support manufacturing and assembly of power components
- Collaborated on the power system for a flywheel trebuchet, integrating a 212cc gas motor into the assembly
- Iterated designs based on team feedback and testing to improve system reliability

Apogee Ceramics Inc. — Mechanical Engineering Co-op *Paris, ON • 2024 – 2025*

- Modeled procedural fixtures and safety guards in SolidWorks, applying DFMA to simplify manufacturing
- Evaluated material and design options using decision-matrix analysis for cost-effective solutions
- Authored and updated SOPs to align with company policy

McMaster Engineering Community Outreach — STEM Outreach Facilitator *Hamilton, ON • 2026*

- Delivered hands-on STEM workshops and engineering demos to K-12 students across the GTA, adapting content for different age groups
- Planned outreach events to boost STEM participation among underrepresented communities